

Blipscope Wi-Fi Field Test

Diagnostic firmware — blipscope-s3-128-joindiag, built from v11. **These boards are test units and must be re-flashed to shipping firmware before going near a customer.**

What you are trying to find out: whether the board that failed here is *broken*, or whether *this network* rejects Blipscopes. One board can't tell you. Four boards, all at the same place, can.

Before you leave

Board (on screen)	What it is	Why it's coming
Blipscope-326E64	THE SUSPECT	The board that actually failed here. Everything hinges on this one.
Blipscope-31E794	KNOWN GOOD	Has joined reliably for weeks. If this one fails too, the network is at fault.
Blipscope-31DF20	New, same batch	Tests whether the whole production batch is affected.
Blipscope-31C2C8	New, same batch	Second sample of the batch.

- **Take:** 4 boards, 4 USB power sources (chargers or a battery pack), this sheet, a pen, a phone.
- **You do NOT need a laptop.** Everything appears on the boards' screens.
- **Each board names itself on screen** in white, under the yellow banner. That's how you tell them apart.

Before powering anything — the 2-minute check that could save the trip.
Look at each board's antenna: the **ANT1/RF1 0-ohm link** and whether the **u.FL connector is fully clicked down**. Photograph all four. If a link is in the wrong position or a u.FL is loose, *that is the answer* — stop and come home.

Ask the owner first

Write these down — or photograph the eero app screens:

eero model	
Firmware version	
How many eero units	
Encryption, 2.4 GHz / 5 GHz	
Network name (SSID)	

RUN A — their network exactly as it is

Do all four boards. Same procedure each time.

1. **Power the board.** Wait for the yellow **** WIFI DIAG **** banner.
2. **On your phone**, open Wi-Fi settings and connect to the hotspot matching the name on that board's screen (e.g. Blipscope-326E64).
3. A setup page opens automatically. If not, browse to 192.168.4.1
4. **Enter their network name and password.** Save.
5. **Now wait 12 minutes.** Leave it alone. See the timing box below.
6. **Photograph the screen** and fill in the row on page 3.

Twelve minutes, and don't cut it short. This is the whole reason the test exists — the original problem was called a failure after 30 seconds.

The board tries for about **1.5 minutes**, shows the setup screen for **3 minutes**, then restarts and tries again. One full round is about **5 minutes**. **You need at least two full rounds** to tell "failed" from "still trying".

The cyc number on screen counts rounds. **Wait until it reaches at least 3.**

Reading the screen

On screen	Means
Blipscope-326E64	Which board this is. Always check this before writing anything down.
cyc 3 up 2m10s try 7	3 rounds done, 2m10s this round, 7 join attempts.
SSID TheirNetwork	The network it is trying. Check it's the right one — a typo looks like a fault.
assoc Y ip Y	It worked. Joined and got an address.
#4 r=201 NO_AP_FOUND	Can't see the network at all.
#4 r=202 AUTH_FAIL	Password rejected. Re-check what you typed.
#4 r=204 4WAY_HANDSHAKE_TIMEOUT	The one we're hunting. It started the handshake and it didn't finish.
ONLY ON 5GHz	Stop — that's the answer. These boards are 2.4 GHz only. Note it and move on.
..38:2A c6/2.4 -61dB	One of their access points: last digits, channel, band, signal. Weaker than about -75 is poor.

A blank or "Connecting to Wi-Fi..." screen is NOT a result. For the first ~1.5 minutes of every round that's normal. If it stays blank for more than 3 minutes, or cyc never changes, something went wrong with the test — note it as "no data", not as a failure.

RUN B — with 5 GHz paused

Only after Run A is finished and written down.

1. Ask the owner to open the **eero app** and go to:
Settings → Troubleshooting → "My device won't connect" → "My device is 2.4 GHz only"
and **pause 5 GHz and 6 GHz**. This lasts 30 minutes.
2. **Note the time you started** — everything must happen inside that window.
3. Power-cycle all four boards (unplug, plug back in).
4. Same as Run A: wait for cyc 3, photograph, record.
5. **Ask the owner to restore the setting** when you're done. Don't leave their network changed.

Record your results

Circle JOINED or FAILED. If failed, write the r= number and name.

Board	Run	Result	cyc	Reason code / notes
326E64 (suspect)	A	JOINED / FAILED		
31E794 (known good)	A	JOINED / FAILED		
31DF20	A	JOINED / FAILED		
31C2C8	A	JOINED / FAILED		
326E64 (suspect)	B	JOINED / FAILED		
31E794 (known good)	B	JOINED / FAILED		
31DF20	B	JOINED / FAILED		
31C2C8	B	JOINED / FAILED		

What the combinations mean

If...	Then
Suspect fails, known-good joins	The board is faulty. Hardware — check its antenna. Question answered.
Suspect + both new boards fail, known-good joins	Batch defect. Affects the other 45 units. Important finding.
All four fail	Their network. The hardware theory is dead.
Run A fails, Run B joins	Band steering. The mesh wasn't giving the board 2.4 GHz.
Suspect joins fine	Intermittent — not a clean bill of health. Note it and run again.

Bring home: photographs of every screen (each one names its own board), photographs of the four antennas, this sheet filled in, and the eero details from page 1.

Photograph screens straight on and close enough to read the small text — the photo is the evidence.